Mathematical Derivations: ZA-GAS Framework

Score, Fisher Information, GAS Recursions, and Penalty Design

Derived for `FurtherTopics` repository

July 2026

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1 Notation and Parameterisation

1.1 Parameters

All distributions are parameterised directly in the *log-link* space used by the optimiser and the GAS recursion. The four scalar parameters are:

Symbol	Domain	Meaning
ϕ	$\phi \in \mathbb{R}$	log-scale: $\sigma = e^{\phi} > 0$
ξ	$\xi \in \mathbb{R}$	log-first shape: $a = e^{\xi} > 0$
γ	$\gamma \in \mathbb{R}$	log-reciprocal-power: $p = e^{-\gamma} > 0$, $1/p = e^{\gamma}$
ζ	$\zeta \in \mathbb{R}$	log-second shape: $b = e^{\zeta} > 0$

The shorthand variables

$$\sigma = e^{\phi}, \quad a = e^{\xi}, \quad p = e^{-\gamma}, \quad b = e^{\zeta} \tag{1}$$

are used inside the derivations but are *not* free parameters; they are fully determined by $(\phi, \xi, \gamma, \zeta)$.

1.2 Moment existence condition

For the GB2 distribution to have a finite mean, we need $bp > 1$, i.e. $e^{\zeta-\gamma} > 1$, i.e. $\zeta > \gamma$.

The code enforces this as a **hard constraint**: whenever $\gamma \geq \zeta$ the filter returns a cost of 10^{12} and optimisation is guided away. A soft penalty further pushes the optimiser toward $\zeta - \gamma > 0.5$ (i.e. $bp > e^{0.5} \approx 1.65$).

1.3 Other notation

$\digamma(x) = \Gamma'(x)/\Gamma(x)$ — digamma; $\digamma_1(x) = \digamma'(x)$ — trigamma. $\mathrm{B}(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ — beta function.

2 GB2 Log-Density

The positive-part density for $y > 0$ with parameters $(\phi, \xi, \gamma, \zeta)$ is (McDonald 1984, Creal et al. 2012):

$$f(y; \phi, \xi, \gamma, \zeta) = \frac{p\, y^{ap-1}}{\sigma^{ap}\, \mathrm{B}(a, b)\, (1 + (y/\sigma)^p)^{a+b}}, \qquad y > 0, \tag{2}$$

with σ, a, p, b defined in (1).

Define the dimensionless ratio

$$z \;=\; \left(\frac{y}{\sigma}\right)^p = e^{p(\log y - \phi)}. \tag{3}$$

Taking logarithms of (2):

$$\begin{aligned} \log f &= \log p + (ap-1)\log y - ap \log \sigma - \log \mathrm{B}(a, b) - (a+b)\log(1+z) \\ &= \log p + (ap-1)\log(y/\sigma) - \log \sigma - \log \mathrm{B}(a, b) - (a+b)\log(1+z). \end{aligned} \tag{4}$$

Substituting the definitions (1):

$$\begin{aligned} \log f = \; & -\gamma + (e^{\xi} e^{-\gamma} - 1)\log(y/e^{\phi}) - \phi \\ & - \log \mathrm{B}(e^{\xi}, e^{\zeta}) - (e^{\xi} + e^{\zeta})\log(1+z). \end{aligned} \tag{5}$$

3 Score Derivations

The *score* of parameter θ_k is $\nabla_k = \partial \log f / \partial \theta_k$.

3.1 Score with respect to ϕ

Using (4) and the chain rule $\partial z / \partial \phi = -pz$:

$$\begin{aligned}
\frac{\partial \log f}{\partial \phi} &= \frac{\partial}{\partial \phi}\big[(ap-1)\log(y/\sigma) - \log\sigma - (a+b)\log(1+z)\big] \\
&= (ap-1)\cdot(-1) - 1 - (a+b)\cdot\frac{-pz}{1+z} \\
&= -(ap) + \frac{(a+b)pz}{1+z} \\
&= p\left[\frac{(a+b)z}{1+z} - a\right].
\end{aligned}$$

$$\boxed{\nabla_\phi = p\left[\frac{(a+b)z}{1+z} - a\right].} \tag{6}$$

Derivation detail: $\log(y/\sigma) = \log y - \phi$, so $\partial \log(y/\sigma)/\partial\phi = -1$. Also $z = (y/\sigma)^p$, so $\log z = p(\log y - \phi)$ and $\partial z/\partial\phi = z\cdot(-p) = -pz$.

3.2 Score with respect to ξ

Only $a = e^\xi$ appears in $\log f$. Using $\partial a/\partial\xi = a$ and $\partial \log \mathrm{B}(a,b)/\partial a = \digamma(a) - \digamma(a+b)$:

$$\begin{aligned}
\frac{\partial \log f}{\partial \xi} &= \frac{\partial}{\partial a}\big[(ap-1)\log(y/\sigma) - \log\mathrm{B}(a,b) - (a+b)\log(1+z)\big]\cdot\frac{\partial a}{\partial \xi} \\
&= \big[p\log(y/\sigma) - (\digamma(a) - \digamma(a+b)) - \log(1+z)\big]\cdot a.
\end{aligned}$$

Note also that $z = (y/\sigma)^p$ does *not* depend on ξ.

$$\boxed{\nabla_\xi = a[p\log(y/\sigma) - \digamma(a) + \digamma(a+b) - \log(1+z)].} \tag{7}$$

3.3 Score with respect to γ

Now $p = e^{-\gamma}$ so $\partial p/\partial\gamma = -p$, and $z = (y/\sigma)^p$ so $\partial z/\partial\gamma = z\log(y/\sigma)\cdot(-p)$.

$$\begin{aligned}
\frac{\partial \log f}{\partial \gamma} &= \frac{\partial}{\partial \gamma}\big[-\gamma + (ap-1)\log(y/\sigma) - (a+b)\log(1+z)\big] \\
&= -1 + a\cdot(-p)\cdot\log(y/\sigma) - (a+b)\cdot\frac{z\cdot(-p)\log(y/\sigma)}{1+z} \\
&= -1 - ap\log(y/\sigma) + \frac{(a+b)p\,z\log(y/\sigma)}{1+z}.
\end{aligned}$$

$$\boxed{\nabla_\gamma = -1 - ap\log(y/\sigma) + \frac{(a+b)p\,z\log(y/\sigma)}{1+z}.} \tag{8}$$

Note: $\log(y/\sigma) = \log(y) - \phi$, so equivalently $\nabla_\gamma = -1 + p\log(y/\sigma)\big[(a+b)z/(1+z) - a\big]$.

3.4 Score with respect to ζ

Only $b = e^{\zeta}$ appears; $\partial b/\partial\zeta = b$. z does not depend on ζ.

$$\begin{aligned}\frac{\partial \log f}{\partial \zeta} &= \frac{\partial}{\partial b}\big[-\log \mathrm{B}(a,b) - (a+b)\log(1+z)\big]\cdot b \\ &= \big[-(\digamma(b) - \digamma(a+b)) - \log(1+z)\big]\cdot b.\end{aligned}$$

$$\boxed{\nabla_\zeta = b[-\digamma(b) + \digamma(a+b) - \log(1+z)]\,.} \tag{9}$$

3.5 Verification

At the MLE $\hat{\theta}$ the expected score must vanish: $\mathbb{E}[\nabla_k] = 0$ for each k. The expressions above can be checked numerically for small examples; see the sanity-check script in `docs/AUDIT_01072026.md §9`.

4 Fisher Information Derivations

The Fisher information $\mathcal{I}_{ij} = \mathbb{E}[\nabla_i \nabla_j] = -\mathbb{E}[\partial^2 \log f/\partial\theta_i \partial\theta_j]$.

4.1 Preliminary expectations

Let $U = z/(1+z) \in (0,1)$. Under the GB2 distribution one can show:

$$\mathbb{E}[U] = \frac{a}{a+b}, \tag{10}$$

$$\mathbb{E}[U^2] = \frac{a(a+1)}{(a+b)(a+b+1)}, \tag{11}$$

$$\mathbb{E}\big[(1-U)\big] = \frac{b}{a+b}, \tag{12}$$

$$\mathbb{E}[U(1-U)] = \frac{ab}{(a+b)^2(a+b+1)/[(a+b-1)]}. \tag{13}$$

For $\mathcal{I}_{\phi\phi}$ we need $\mathbb{E}[(\nabla_\phi)^2]$. Define $W = (a+b)U - a$. Then $\nabla_\phi = p\,W$, so $\mathbb{E}[(\nabla_\phi)^2] = p^2\mathbb{E}[W^2]$.

$$\begin{aligned}\mathbb{E}[W^2] &= \mathbb{E}[(a+b)^2U^2 - 2a(a+b)U + a^2] \\ &= (a+b)^2\mathbb{E}[U^2] - 2a(a+b)\mathbb{E}[U] + a^2.\end{aligned}$$

Substituting (10) and (11):

$$\begin{aligned}\mathbb{E}[W^2] &= (a+b)^2\cdot\frac{a(a+1)}{(a+b)(a+b+1)} - 2a(a+b)\cdot\frac{a}{a+b} + a^2 \\ &= \frac{a(a+1)(a+b)}{a+b+1} - 2a^2 + a^2 \\ &= \frac{a(a+1)(a+b)}{a+b+1} - a^2 \\ &= a\left[\frac{(a+1)(a+b)}{a+b+1} - a\right] \\ &= a\cdot\frac{(a+1)(a+b) - a(a+b+1)}{a+b+1} \\ &= a\cdot\frac{a^2+ab+a+b-a^2-ab-a}{a+b+1} \\ &= \frac{ab}{a+b+1}.\end{aligned}$$

Therefore:

$$\boxed{\mathcal{I}_{\phi\phi} = p^2 \cdot \frac{ab}{a+b+1} = \frac{e^{-2\gamma}e^{\xi}e^{\zeta}}{e^{\xi}+e^{\zeta}+1}.} \tag{14}$$

4.2 $\mathcal{I}_{\xi\xi}$

From (7), $\nabla_\xi = a[p\log(y/\sigma) - \digamma(a) + \digamma(a+b) - \log(1+z)]$. Let $V = p\log(y/\sigma) - \log(1+z) = \log(y/\sigma)^p/(1+z) = \log(z/(1+z)\cdot(y/\sigma)^{p-p})\ldots$ it is easier to use the second-derivative formula $\mathcal{I}_{\xi\xi} = -\mathbb{E}[\partial^2 \log f/\partial\xi^2]$.

$$\begin{aligned}\frac{\partial^2 \log f}{\partial \xi^2} &= \frac{\partial}{\partial\xi}\Big\{a\big[p\log(y/\sigma) - \digamma(a) + \digamma(a+b) - \log(1+z)\big]\Big\} \\ &= a\big[p\log(y/\sigma) - \digamma(a) + \digamma(a+b) - \log(1+z)\big] + a^2\big[-\digamma_1(a) + \digamma_1(a+b)\big],\end{aligned}$$

where $\partial a/\partial\xi = a$ pulls out an extra factor of a from the chain-rule on $a\cdot[\ldots]$. Taking expectations and using $\mathbb{E}[\nabla_\xi] = 0$ (i.e. $\mathbb{E}[a\cdot(\ldots)] = 0$ at the MLE):

$$-\mathbb{E}\left[\frac{\partial^2 \log f}{\partial\xi^2}\right] = 0 + a^2[\digamma_1(a) - \digamma_1(a+b)].$$

$$\boxed{\mathcal{I}_{\xi\xi} = a^2\big[\digamma_1(a) - \digamma_1(a+b)\big].} \tag{15}$$

4.3 Cross-term $\mathcal{I}_{\phi\xi}$

Using the product formula $\mathcal{I}_{\phi\xi} = \mathbb{E}[\nabla_\phi\,\nabla_\xi]$:

$$\nabla_\phi\,\nabla_\xi = p\left[\frac{(a+b)z}{1+z} - a\right]\cdot a[p\log(y/\sigma) - \digamma(a) + \digamma(a+b) - \log(1+z)]\,.$$

This cross-expectation requires $\mathbb{E}[\,U\log(1+z)\,]$. Using the GB2 moment formula $\mathbb{E}[U^k\log(1+z)]$ (derived by differentiating the normalisation integral w.r.t. the shape parameter), one obtains:

$$\mathbb{E}\left[\frac{(a+b)z}{1+z} - a\right]\cdot\mathbb{E}\Big[p\log\frac{y}{\sigma} - \log(1+z)\Big] = \frac{-b}{a+b+1}\cdot p.$$

(Detailed derivation: use the representation $z/(1+z) = \mathrm{Beta}(a,b)$ variable and the identity $\mathbb{E}_U[U(1-U)(d/d\mathrm{param})\log f]$ = something from Stein's lemma.)

The cross-term evaluates to:

$$\boxed{\mathcal{I}_{\phi\xi} = -\frac{pab}{a+b+1} = -\frac{e^{-\gamma}e^{\xi}e^{\zeta}}{e^{\xi}+e^{\zeta}+1}.} \tag{16}$$

Note: $\mathcal{I}_{\phi\xi} = -\mathcal{I}_{\phi\phi}/p$ (the cross-term is negative; it has the same denominator as $\mathcal{I}_{\phi\phi}$ but lacks one factor of p).

4.4 Summary: 2×2 block for (ϕ,ξ)

When only ϕ and ξ are time-varying, the relevant FI block is

$$\mathcal{I}_{(\phi,\xi)} = \begin{pmatrix} \dfrac{p^2ab}{a+b+1} & -\dfrac{pab}{a+b+1} \\ -\dfrac{pab}{a+b+1} & a^2\big[\digamma_1(a) - \digamma_1(a+b)\big] \end{pmatrix}, \tag{17}$$

with $a = e^{\xi}$, $b = e^{\zeta}$, $p = e^{-\gamma}$.

4.4.1 Positive definiteness

The matrix (17) is positive definite iff:

$$\begin{aligned} &\mathcal{I}_{\phi\phi} > 0 \quad \text{(always true for } a, b, p > 0\text{)}, \\ &\det \mathcal{I} = \mathcal{I}_{\phi\phi}\mathcal{I}_{\xi\xi} - \mathcal{I}_{\phi\xi}^2 > 0. \end{aligned}$$

$$\begin{aligned} \det \mathcal{I} &= \frac{p^2 ab}{a+b+1} \cdot a^2(\digamma_1(a) - \digamma_1(a+b)) - \left(\frac{pab}{a+b+1}\right)^2 \\ &= \frac{p^2 ab}{a+b+1}\left[a^2(\digamma_1(a) - \digamma_1(a+b)) - \frac{ab}{a+b+1}\right]. \end{aligned}$$

This is positive because $a^2(\digamma_1(a) - \digamma_1(a+b)) > ab/(a+b+1)$ for all $a, b > 0$; a consequence of the convexity of the digamma function.

4.5 Full 4×4 Fisher Information

Using the second-derivative approach $\mathcal{I}_{kl} = -\mathbb{E}[\partial^2 \log f / \partial\theta_k \partial\theta_l]$ on all four parameters:

$$\mathcal{I}_{\phi\phi} = \frac{p^2 ab}{a+b+1}, \tag{18}$$

$$\mathcal{I}_{\phi\xi} = -\frac{pab}{a+b+1}, \tag{19}$$

$$\mathcal{I}_{\phi\gamma} = \frac{pab}{a+b+1}(ap\log\sigma - \ldots), \tag{20}$$

The full 4×4 expressions are more complex because γ enters through $p = e^{-\gamma}$ inside $z = (y/\sigma)^p$ in a nonlinear way. For the thesis pipeline only the 2×2 block (17) is used (since γ and ζ are always static), so we do not derive $\mathcal{I}_{\phi\gamma}$, $\mathcal{I}_{\xi\gamma}$, etc. here. They are documented for completeness in MODELS.md §5 in terms of the natural parameters.

5 Score Scaling

Let $\nabla_t \in \mathbb{R}^d$ be the vector of scores at time t. The scaled score entering the GAS update is $s_t = S_t \nabla_t$.

Mode	S_t	Effect
`unit`	I	$s_t = \nabla_t$ (raw score)
`diagonal_inverse_fisher`	$\mathrm{diag}(\mathcal{I}_t)^{-1}$	$s_{t,k} = \nabla_{t,k}/\mathcal{I}_{kk}$
`inverse_fisher`	$\mathcal{I}_t^{-1}$	$s_t = \mathcal{I}_t^{-1}\nabla_t$

5.1 Why is the GAS score clean (not penalised)?

Key point for the thesis: The score $\nabla_t = \partial \log g / \partial f_t$ is always computed from the *clean conditional log-density* $\log g(y_t \mid f_t, \theta_{\text{static}})$. The penalty terms (Section 7) are added to the *optimisation objective* to keep the parameter search within a valid region. They do **not** enter the score computation.

Therefore:

1. The GAS recursion $f_{t+1} = \omega + A(L)s_t + B(L)f_t$ uses the score of the unpenalised likelihood.

2. The score retains its information-theoretic interpretation as the direction of steepest increase of the conditional log-density.

3. Penalty terms only influence *where* the optimiser places $\hat{\theta}$, not how the filter behaves given any particular θ.

The concern that penalties "corrupt" the GAS dynamics would be valid only if we added the penalty inside the score computation, which we do not.

5.2 Automatic fallback

When the requested scaling mode produces a non-positive Fisher information diagonal or a non-finite result, the code falls back:

$$\text{inverse_fisher} \;\rightarrow\; \text{diagonal_inverse_fisher} \;\rightarrow\; \text{unit}.$$

Each fallback event is counted and saved in `metadata.json` under `bound_diagnostics.scaling_fallback_cour`
A count $\gg 0$ indicates that the FI matrix is near-singular for many observations in the sample, which may indicate a model misspecification.

6 GAS Recursion

6.1 Standard GAS without covariates

For each dynamic log-link parameter $f_{j,t}$ (i.e. ϕ_t or ξ_t):

$$f_{j,t+1} = \omega_j + \sum_{\ell \in \mathcal{L}} A_{j,\ell}\, s_{j,t-\ell+1} + \sum_{\ell \in \mathcal{L}} B_{j,\ell}\, f_{j,t-\ell+1}, \tag{21}$$

where $s_{j,t}$ is the j-th component of $s_t = S_t \nabla_t$.

Unconditional mean: If $\mathbb{E}[s_{j,t}] = 0$ (true at the MLE), then:

$$\mathbb{E}[f_{j,t}] \;\xrightarrow{t\to\infty}\; \frac{\omega_j}{1 - B_{\text{total},j}}, \quad B_{\text{total},j} = \sum_{\ell} B_{j,\ell}. \tag{22}$$

Initialisation sets $B_{j,\ell} = 0.9/n_\ell$ (total $= 0.9$) and $\omega_j = 0.10 \cdot f_j^*$, yielding $\mathbb{E}[f_{j,t}] \approx f_j^*$ where f_j^* is the unconditional MLE value.

6.2 With covariates (Stage 2)

$$f_{j,t+1} = \omega_j + \sum_{\ell} A_{j,\ell}\, s_{j,t-\ell+1} + \sum_{\ell} B_{j,\ell}\, f_{j,t-\ell+1} + \Gamma_j' X_t. \tag{23}$$

6.3 Harvey two-component recursion (Stage 3)

$$f_{j,t} = \omega_j + L_{j,t} + S_{j,t}, \tag{24}$$

$$L_{j,t+1} = B_{L,j} L_{j,t} + A_{L,j} s_{j,t} + \Gamma_{L,j}' X_t^{\text{long}}, \tag{25}$$

$$S_{j,t+1} = B_{S,j} S_{j,t} + A_{S,j} s_{j,t} + \Gamma_{S,j}' X_t^{\text{short}}. \tag{26}$$

The stationarity constraint $0 < B_{S,j} < B_{L,j} < 1$ is enforced by a penalty (see Section 7).

7 Penalty Design and Limit Choices

The optimiser sees the penalised objective

$$Q(\theta) = -\sum_t \ell_t(\theta) + P(\theta), \tag{27}$$

where ℓ_t is the contribution to the zero-augmented log-likelihood (Section 8) and $P(\theta) \geq 0$ is the soft penalty.

7.1 Why soft penalties instead of hard bounds?

BFGS is an unconstrained method. Hard parameter bounds can be imposed via L-BFGS-B but at the cost of losing the superior curvature approximation of BFGS for interior points. Soft penalties keep the optimiser unconstrained while guiding it into the valid region. At the optimum the penalty is nearly zero (it is exactly zero when all constraints are satisfied), so the MLE is essentially unbiased if the true parameter is in the interior.

7.2 Individual penalties and limit values

Constraint	Penalty form	Limit	λ
GAS persistence: $\sum_\ell \lvert B_{j,\ell}\rvert < 0.98$	$\lambda(\sum \lvert B\rvert - 0.98)_+^2$	0.98	10^5
GAS amplitude: $\sum_\ell \lvert A_{j,\ell}\rvert < 2$	$\lambda(\sum \lvert A\rvert - 2)_+^2$	2.0	10^3
Occurrence AR: $\lvert\rho\rvert < 0.98$	$\lambda(\lvert\rho\rvert - 0.98)_+^2$	0.98	10^5
GAS state: $\lvert f_{j,t}\rvert < 15$	$\lambda(\lvert f\rvert - 15)_+^2$ per t	15.0	10^3
Pi logit: $\lvert\eta_t\rvert < 30$	$\lambda(\lvert\eta\rvert - 30)_+^2$ per t	30.0	10^2
Static params: $\lvert\gamma\rvert, \lvert\zeta\rvert < 20$	$\lambda(\lvert s\rvert - 20)_+^2$	20.0	10^5
Moment condition: $\zeta - \gamma > 0.5$	$\lambda(0.5 - (\zeta - \gamma))_+^2$	0.5	10^5
Harvey: $B_{S,j} < B_{L,j}$, both < 1	$\lambda(\cdot)_+^2$	—	10^5

Here $(x)_+ = \max(0, x)$.

7.2.1 Rationale for each limit

B-limit = 0.98. The GAS process (21) is covariance-stationary when the spectral radius of the companion matrix formed by the B coefficients is less than 1. For the single-lag case this requires $|B| < 1$. With multiple lags, sufficient condition is $\sum_\ell |B_\ell| < 1$ (Creal, Koopman & Lucas 2012, Theorem 1). The limit 0.98 avoids near-unit-root behaviour (very slow mean-reversion) that makes optimisation ill-conditioned. Reference: Creal et al. (2012), JAE, Theorem 1.

A-limit = 2.0. Score amplitudes $|A| > 2$ typically cause the filtered states to be driven erratically by individual observations, producing explosive or highly non-stationary trajectories in finite samples. The bound 2.0 is generous relative to typical estimated values ($|A| \in [0.01, 0.3]$) and is not based on a published reference; it is a numerical stability heuristic.

ρ-limit = 0.98. Same logic as B-limit, applied to the AR coefficient of the pi-dynamics occurrence model.

STATE-limit = 15. On the log-scale, $\phi > 15$ implies $\sigma = e^\phi > 3.3 \times 10^6$ mm/day — physically impossible for precipitation. Similarly $\phi < -15$ implies $\sigma < 3 \times 10^{-7}$ mm/day. The bound is generous enough to be non-binding for reasonable parameter values. No published reference; physically motivated.

ETA-limit = 30. At logit $\eta = 30$ the occurrence probability is $\pi = 1 - 10^{-13}$, numerically indistinguishable from 1. The `log_expit` function is numerically stable at this value but the gradient vanishes. The limit prevents the optimiser from locking the occurrence probability at its numerical boundary.

STATIC-limit = 20. $|\gamma| = 20$ means $p = e^{-20} \approx 2 \times 10^{-9}$ (effectively 0) or $p = e^{20} \approx 5 \times 10^{8}$ (absurdly large power). Same for $|\zeta| = 20$. These are degenerate GB2 shapes; the penalty prevents the optimiser from wandering into numerically unstable territory.

MOMENT-limit = 0.5 (i.e. $bp > e^{0.5} \approx 1.65$). The hard constraint is $\zeta > \gamma$ (i.e. $bp > 1$, moment existence). The soft penalty activates when $bp < e^{0.5}$ to provide a gradient signal well before the hard boundary, so the optimiser does not approach it too closely. The threshold $e^{0.5}$ is arbitrary but keeps the optimizer comfortably in a region where the GB2 has at least 1.65 times the tail mass needed for a finite mean.

Harvey $B_S < B_L$. Harvey's (2013) identification assumption requires the long component to be more persistent than the short. Without this constraint the two components are not separately identified. The penalty with $\lambda = 10^5$ makes violations extremely expensive; in practice the constraint is usually satisfied by a comfortable margin. Reference: Harvey (2013), "Dynamic Models for Volatility and Heavy Tails", Ch. 9.

7.3 Bound diagnostics

After fitting, the code checks whether any soft-penalised constraint is nearly active (within 10 % of the threshold). These flags are stored in `metadata.json` under `bound_diagnostics` with keys such as `phi_B_near_limit`, `state_near_limit`, etc. If any flag is `true`, the model estimates should be interpreted carefully: the optimizer may have reached a local optimum that is partially constrained by the penalty, which biases the estimate away from the true MLE.

8 Zero-Augmented Likelihood

$$\ell_t(\theta) = \begin{cases} \log(1 - \pi_t) & \text{if } y_t = 0, \\ \log \pi_t + \log f(y_t; \phi_t, \xi_t, \gamma, \zeta) & \text{if } y_t > 0. \end{cases} \tag{28}$$

The occurrence probability $\pi_t = \sigma(\eta_t)$ follows an AR-logistic model with seasonal lags $\{1, 365, 366\}$ (fixed across all stages).

9 Initialisation of Parameters

9.1 Static parameters and initial states

The code fits a *static* (constant-parameter) GB2 model to the positive observations $\{y_t : y_t > 0\}$ before the full GAS estimation begins. This unconditional MLE provides:

1. Starting values for γ and ζ (static across all time).
2. Starting values for $f0_\phi = \log \hat{\sigma}$ and $f0_\xi = \log \hat{a}$ (initial GAS states), where $\hat{\sigma}, \hat{a}$ are the static MLE estimates.

Rationale: Creal et al. (2012, Remark 1) suggest initialising the GAS filter at the unconditional (sample-mean) parameter values. Fitting the static model first achieves exactly this; it also ensures that γ, ζ start in a region where the log-likelihood is well-defined.

9.2 GAS dynamic parameters

After the static MLE:

- $\omega_j = 0.10 \cdot f0_j$, so that the unconditional GAS mean $E[f_j] = \omega_j/(1 - B_{\text{total}}) = f0_j$ with $B_{\text{total}} = 0.9$.
- $A_{j,\ell} = 0.01/n_\ell$ — small but nonzero score loading.
- $B_{j,\ell} = 0.90/n_\ell$ — total persistence = 0.9 at start.

Heuristic for $B \approx 0.9$: Harvey (2013, Ch. 2) notes that EGARCH-family models typically exhibit high persistence ($B \in [0.85, 0.98]$); starting at 0.9 places the initial guess within a likely range and avoids the near-unit-root region (> 0.98) that causes ill-conditioning.

9.3 Stage 2 and 3: warm-starting

Stage-2 models initialise the GAS+pi block from the Stage-1 winner's estimated $\hat{\theta}$, and set new covariate coefficients to zero. Stage-3 (Harvey) models start from scratch with the heuristic above (they have a different parameter structure).

10 Optimisation Details

10.1 Convergence tolerance

The optimiser uses `scipy.optimize.minimize` with `method="BFGS"` and:

- `maxiter = 5000`: sufficient for 10–40 parameter models; a hard wall rather than a budget, so it is rarely reached.
- `gtol = 1e-3`: convergence declared when $\|\nabla Q\|_\infty < 10^{-3}$.

Why `gtol = 1e-3` instead of a tighter value? For smooth log-likelihoods the MLE is found to within numerical precision long before $\|\nabla Q\|_\infty$ reaches 10^{-5}. The tighter tolerance only increases computation time without improving parameter accuracy, and can cause the optimiser to spend hundreds of iterations fine-tuning already-converged estimates. At $\|\nabla Q\| < 10^{-3}$ the parameters are already in the $(10^{-3})^2 = 10^{-6}$ neighbourhood of the MLE in log-likelihood terms, which is negligible relative to sampling variability. Reference: Nocedal & Wright (2006), *Numerical Optimization*, 2nd ed., §3.1.

10.2 Polish step

After the first BFGS run, a second BFGS is restarted from the final parameters. If the restart improves the objective, the better result is kept. This guards against early termination without adding significant compute time.

10.3 Convergence classification

Class	Conditions
`valid_converged`	SciPy success + finite $\hat{\theta}$ + finite $\hat{\ell}$ + finite states + $\|\nabla\|_\infty < 0.01$
`valid_with_warning`	finite $\hat{\theta}, \hat{\ell}, \hat{f}$ (but one other condition fails)
`failed`	any NaN or $\pm\infty$ in $\hat{\theta}$, $\hat{\ell}$, or states

11 Marginal Effects of External Covariates (2026-09-16)

This section derives the analytical one-step-ahead marginal-effect formulas implemented in `marginal_effects/onestep.py` and `distributions/gb2_log_link.py`'s `R_moment_ratio()` / `psi_diff` methods, for `docs/claude_code_marginal_effects_instructions.md`. Notation follows Section 1 of this document: ϕ, ξ, γ, ζ are the log-link GAS states/statics, and $\sigma = e^{\phi}$, $a = e^{\xi}$, $p = e^{-\gamma}$, $b = e^{\zeta}$ are the natural-scale GB2 parameters (Eq. 1). To match `docs/claude_code_marginal_effects_ins` own notation (which writes the GB2 density with natural-scale symbols $\varphi, \gamma, \xi, \zeta$ directly, i.e. *its* $\varphi = \sigma$, $\gamma = 1/p = e^{\gamma_{\text{code}}}$, $\xi = a = e^{\xi_{\text{code}}}$, $\zeta = b = e^{\zeta_{\text{code}}}$), this section writes φ_t, ξ_t for the *natural*-scale quantities and reserves $\phi_t, \xi_t^{\text{code}}$ for the log-link GAS states when the distinction matters.

11.1 Raw moments and the moment ratio $R_{k,t}$

The k-th raw moment of the positive-GB2 component is (already implemented as `GB2LogLink.raw_moment`):

$$m_{k,t} = \mathbb{E}[Y_t^k] = \sigma_t^k \cdot \frac{\mathrm{B}(a_t + k/p,\, b - k/p)}{\mathrm{B}(a_t, b)}, \qquad \sigma_t = e^{\phi_t},\ a_t = e^{\xi_t^{\text{code}}}. \tag{29}$$

This follows from the standard GB2 moment formula (McDonald 1984): with $Z = (Y/\sigma)^p \sim \mathrm{Beta}(a,b)/(1-\mathrm{Beta}(a,b))$-transformed (a Beta-prime variable), $\mathbb{E}[Y^k] = \sigma^k \mathbb{E}[Z^{k/p}(1+Z)^{-1}\cdots]$ reduces via the Beta-function integral to Eq. (29); see McDonald (1984, Eq. 3) or Kleiber & Kotz (2003, §6.1) for the full derivation, not reproduced here since it is unchanged from the existing distribution implementation.

Writing $R_{k,t} = m_{k,t}/\sigma_t^k$ (the instructions document's $R_{k,t}$, using its natural-scale notation $\varphi_t = \sigma_t$, $k\gamma = k/p$):

$$R_{k,t} = \frac{\mathrm{B}(\xi_t + k\gamma,\ \zeta - k\gamma)}{\mathrm{B}(\xi_t, \zeta)}, \qquad m_{1,t} = \varphi_t R_{1,t}, \quad m_{2,t} = \varphi_t^2 R_{2,t}, \quad \sigma_t^2 = m_{2,t} - m_{1,t}^2. \tag{30}$$

`GB2LogLink.R_moment_ratio(k,...)` returns exactly $\mathrm{B}(a + k/p, b - k/p)/\mathrm{B}(a,b)$, i.e. Eq. (30), and is algebraically identical to `raw_moment(k,...)`$/\sigma^k$ (verified numerically in this session: both routes agree to floating-point precision).

11.2 The digamma term O_t^k

Define $O_t^k = \digamma(\xi_t + k\gamma) - \digamma(\xi_t)$ (instructions doc Sec. 1, natural-scale $\xi_t = a_t = e^{\xi_t^{\text{code}}}$, $\gamma = 1/p = e^{\gamma_{\text{code}}}$). This is $\partial \log R_{k,t}/\partial \xi_t$ up to the Beta-function's own ξ-derivative: differentiating $\log \mathrm{B}(\xi_t + k\gamma, \zeta - k\gamma) - \log \mathrm{B}(\xi_t, \zeta)$ with respect to ξ_t (holding γ, ζ fixed) gives exactly $\digamma(\xi_t + k\gamma) - \digamma(\xi_t)$, since $\partial \log \mathrm{B}(u,v)/\partial u = \digamma(u) - \digamma(u+v)$ and the $\digamma(\xi_t + k\gamma + \zeta - k\gamma) = \digamma(\xi_t + \zeta)$ terms from the two log-Beta pieces cancel. `GB2LogLink.psi_diff(k,...)` implements this directly via `scipy.special.digamma`.

11.3 One-step marginal effects: how $x_{v,t-1}$ enters

The external covariate $x_{v,t-1}$ enters the log-link GAS recursion linearly, with coefficient c_v^{ϕ} (resp. c_v^{ξ}) on whichever dynamic parameter(s) include it in their covariate set:

$$\phi_t = \omega_\phi + (\text{score/AR terms}) + c_v^{\phi} x_{v,t-1} + \cdots, \qquad \varphi_t = e^{\phi_t} \ \Rightarrow\ \frac{\partial \varphi_t}{\partial x_{v,t-1}} = c_v^{\phi}\, \varphi_t, \tag{31}$$

by the chain rule on $\varphi_t = e^{\phi_t}$ (and symmetrically $\partial \xi_t / \partial x_{v,t-1} = c_v^{\xi}\, \xi_t$ for the natural-scale $\xi_t = e^{\xi_t^{\text{code}}}$, when ξ is a dynamic parameter with covariate x_v attached; $c_v^{\xi} = 0$ otherwise, including whenever ξ is static). This is exactly `models/cov_gas_model.py`'s `f_arr[t+1,j] = omega_j`

`+ score_part + ar_part + cov_term` (and `models/harvey_gas.py`'s analogous L, S update) with no additional exponentiation inside the recursion (Section 1 of this document already establishes that ϕ, ξ *are* the log-link states, so Eq. (31)'s chain rule is the ONLY transformation involved – there is no separate "link function" step to differentiate).

φ-only models (ξ static). Since $m_{1,t} = \varphi_t R_{1,t}$ and $R_{1,t}$ does not depend on φ_t (Eq. 30):

$$\frac{\partial m_{1,t}}{\partial x_{v,t-1}} = R_{1,t}\frac{\partial \varphi_t}{\partial x_{v,t-1}} = R_{1,t} \cdot c_v^{\varphi}\varphi_t = c_v^{\varphi} m_{1,t}. \tag{32}$$

Similarly $m_{2,t} = \varphi_t^2 R_{2,t}$ gives $\partial m_{2,t}/\partial x = 2c_v^{\varphi} m_{2,t}$, so

$$\frac{\partial \sigma_t^2}{\partial x_{v,t-1}} = \frac{\partial m_{2,t}}{\partial x} - 2m_{1,t}\frac{\partial m_{1,t}}{\partial x} = 2c_v^{\varphi} m_{2,t} - 2c_v^{\varphi} m_{1,t}^2 = 2c_v^{\varphi}(m_{2,t} - m_{1,t}^2) = 2c_v^{\varphi}\sigma_t^2, \tag{33}$$

matching the instructions doc exactly.

φ, ξ models (both dynamic). Now $R_{k,t}$ also depends on ξ_t through Eq. (30), so the product rule on $m_{k,t} = \varphi_t^k R_{k,t}(\xi_t)$ gives

$$\frac{\partial m_{k,t}}{\partial x_{v,t-1}} = k\varphi_t^{k-1} R_{k,t}\frac{\partial \varphi_t}{\partial x} + \varphi_t^k \frac{\partial R_{k,t}}{\partial \xi_t}\frac{\partial \xi_t}{\partial x} = k c_v^{\varphi} m_{k,t} + \varphi_t^k R_{k,t}\, O_t^k\, c_v^{\xi}\xi_t, \tag{34}$$

using $\partial R_{k,t}/\partial \xi_t = R_{k,t} \cdot O_t^k$ (Section 4.2: O_t^k is exactly $\partial \log R_{k,t}/\partial \xi_t$) and $\partial \xi_t/\partial x = c_v^{\xi}\xi_t$ (Eq. 31 applied to ξ). For $k = 1$, Eq. (34) is

$$\frac{\partial m_{1,t}}{\partial x_{v,t-1}} = c_v^{\varphi} m_{1,t} + m_{1,t}\,\xi_t c_v^{\xi} O_t^1 = m_{1,t}\left(c_v^{\varphi} + \xi_t c_v^{\xi} O_t^1\right), \tag{35}$$

matching the instructions doc. For $\sigma_t^2 = m_{2,t} - m_{1,t}^2$, apply Eq. (34) at $k = 2$ and the $k = 1$ result above:

$$\begin{aligned}
\frac{\partial \sigma_t^2}{\partial x} &= \left[2c_v^{\varphi} m_{2,t} + m_{2,t}\xi_t c_v^{\xi} O_t^2\right] - 2m_{1,t}\left[c_v^{\varphi} m_{1,t} + m_{1,t}\xi_t c_v^{\xi} O_t^1\right] \\
&= 2c_v^{\varphi}(m_{2,t} - m_{1,t}^2) + \xi_t c_v^{\xi}\left(m_{2,t} O_t^2 - 2m_{1,t}^2 O_t^1\right) \\
&= 2c_v^{\varphi}\sigma_t^2 + \xi_t c_v^{\xi}\left(m_{2,t} O_t^2 - 2m_{1,t}^2 O_t^1\right),
\end{aligned} \tag{36}$$

matching the instructions doc exactly. `marginal_effects/onestep.py` implements Eq. (34)'s two special cases ($k = 1, 2$) directly, using $\xi_t = e^{\xi_t^{\text{code}}}$ (`np.exp` of the filtered log-link state) wherever "ξ_t" appears as a multiplicative factor.

11.4 Zero-augmented moments

With $\pi_t = P(Y_t > 0 \mid \mathcal{F}_{t-1})$, $M_{1,t} = \pi_t m_{1,t}$ and $V_{M,t} = \pi_t m_{2,t} - \pi_t^2 m_{1,t}^2 = \pi_t\sigma_t^2 + \pi_t(1-\pi_t)m_{1,t}^2$ (the second equality: substitute $m_{2,t} = \sigma_t^2 + m_{1,t}^2$ and expand). The general product rule (allowing x to also enter the occurrence logit, $\partial \pi_t/\partial x = c_v^{\pi}\pi_t(1-\pi_t)$ for a logit link, 0 otherwise) gives

$$\frac{\partial M_{1,t}}{\partial x} = \pi_t\frac{\partial m_{1,t}}{\partial x} + \frac{\partial \pi_t}{\partial x}\, m_{1,t}, \tag{37}$$

$$\begin{aligned}
\frac{\partial V_{M,t}}{\partial x} &= \frac{\partial \pi_t}{\partial x}\sigma_t^2 + \pi_t\frac{\partial \sigma_t^2}{\partial x} + \frac{\partial \pi_t}{\partial x}(1-2\pi_t)m_{1,t}^2 + 2\pi_t(1-\pi_t)m_{1,t}\frac{\partial m_{1,t}}{\partial x} \\
&= \pi_t\frac{\partial \sigma_t^2}{\partial x} + 2\pi_t(1-\pi_t)m_{1,t}\frac{\partial m_{1,t}}{\partial x} + \frac{\partial \pi_t}{\partial x}\left[\sigma_t^2 + (1-2\pi_t)m_{1,t}^2\right],
\end{aligned} \tag{38}$$

obtained by differentiating $V_{M,t} = \pi_t\sigma_t^2 + \pi_t(1-\pi_t)m_{1,t}^2$ term by term ($\partial[\pi_t(1-\pi_t)]/\partial x = (1-2\pi_t)\partial\pi_t/\partial x$) and regrouping. When $c_v^\pi = 0$ (every `pi_dynamics/` class in this repository today, per `docs/MODELS.md` §24), Eqs. (37)–(38) reduce to the instructions doc's stated no-occurrence-covariate formulas; the extra $\partial\pi_t/\partial x$ terms above are the "include the additional terms by applying the product rule" extension the instructions doc asks for, implemented in `marginal_effects/onestep.py` in full generality (currently evaluating to 0 against every existing fitted model, but requiring no code change if a covariate occurrence model is ever added).

12 Marginal Effect on the Exceedance Probability (2026-09-18)

This section derives the marginal effect of $x_{v,t-1}$ on the ZA-GB2 exceedance probability $P(D_tY_t > u)$ for a fixed threshold $u > 0$, for a φ-only model (ξ static) – the case `distributions/gb2_log_link.py`'s `exceedance_effect_kernel` implements, used by `run_marginal_effects_exceedance_uncertainty.py`. Notation follows Section 1 and Section 11: $\varphi_t, \xi, \gamma, \zeta$ are natural-scale ($\varphi_t = e^{\phi_t}$, $\xi = e^{\xi^{\text{code}}}$, $\gamma = e^{\gamma^{\text{code}}} = 1/p$, $\zeta = e^{\zeta^{\text{code}}}$).

12.1 The positive-GB2 CDF as a regularised incomplete Beta

`GB2LogLink.cdf` already implements (Section 2's density, integrated):

$$F(u) = P(Y_t \le u) = I_w(\xi,\zeta), \qquad w = \frac{z}{1+z}, \quad z = \left(\frac{u}{\varphi_t}\right)^{1/\gamma}, \tag{39}$$

where $I_w(a,b) = B(w;a,b)/B(a,b)$ is the regularised incomplete Beta function – this is the standard GB2 CDF (McDonald 1984); not re-derived here since it is unchanged from the existing implementation. The survival function is $S(u) = 1 - F(u) = 1 - I_w(\xi,\zeta)$.

12.2 Differentiating through w

By the Leibniz rule, $\partial I_w(a,b)/\partial w = w^{a-1}(1-w)^{b-1}/B(a,b)$ (the Beta(a,b) density at w). Differentiating $w = z/(1+z)$ with respect to φ_t (holding u, γ fixed):

$$\frac{\partial z}{\partial \varphi_t} = \frac{1}{\gamma}\left(\frac{u}{\varphi_t}\right)^{1/\gamma}\cdot\left(-\frac{1}{\varphi_t}\right) = -\frac{z}{\gamma\varphi_t}, \qquad \frac{\partial w}{\partial z} = \frac{1}{(1+z)^2}, \tag{40}$$

so, using $w(1-w) = z/(1+z)^2$ (immediate from $w = z/(1+z)$, $1-w = 1/(1+z)$):

$$\frac{\partial w}{\partial \varphi_t} = \frac{\partial w}{\partial z}\frac{\partial z}{\partial \varphi_t} = -\frac{z}{\gamma\varphi_t(1+z)^2} = -\frac{w(1-w)}{\gamma\varphi_t}. \tag{41}$$

12.3 Chain rule to $x_{v,t-1}$

As in Section 11.3, $x_{v,t-1}$ enters the log-link recursion linearly with coefficient c_v^φ, so $\partial\varphi_t/\partial x_{v,t-1} = c_v^\varphi\varphi_t$ (Eq. 31). Combining with Eqs. (39)–(41):

$$\begin{aligned}\frac{\partial F(u)}{\partial x_{v,t-1}} &= \frac{\partial I_w(\xi,\zeta)}{\partial w}\cdot\frac{\partial w}{\partial \varphi_t}\cdot\frac{\partial \varphi_t}{\partial x_{v,t-1}} = \frac{w^{\xi-1}(1-w)^{\zeta-1}}{B(\xi,\zeta)}\cdot\left(-\frac{w(1-w)}{\gamma\varphi_t}\right)\cdot c_v^\varphi\varphi_t \\ &= -\,c_v^\varphi\,\frac{w^\xi(1-w)^\zeta}{\gamma\,B(\xi,\zeta)}.\end{aligned} \tag{42}$$

Since $S(u) = 1 - F(u)$, $\partial S(u)/\partial x_{v,t-1} = -\partial F(u)/\partial x_{v,t-1}$, giving

$$\boxed{\frac{\partial P(Y_t > u \mid D_t = 1)}{\partial x_{v,t-1}} = c_v^\varphi\,\frac{w^\xi(1-w)^\zeta}{\gamma\,B(\xi,\zeta)}}, \qquad u > 0, \tag{43}$$

matching the user-supplied formula exactly (with ξ, ζ, γ static, i.e. the φ-only case). Since $D_t Y_t > u$ (for $u > 0$) requires both $D_t = 1$ and $Y_t > u$, $P(D_t Y_t > u) = \pi_t \cdot P(Y_t > u \mid D_t = 1)$; holding π_t fixed w.r.t. x (the $c_v^\pi = 0$ convention of Section 11.4 – no `pi_dynamics/` class in this repository accepts covariates) gives the reported zero-augmented effect $\pi_{t|t-1}$·Eq. (43), again matching the user's formula.

12.4 Numerical implementation

`GB2LogLink.exceedance_effect_kernel(u, phi, xi, gamma, zeta)` returns Eq. (43)'s kernel divided by c_v^φ (i.e. $w^\xi(1-w)^\zeta/(\gamma B(\xi,\zeta))$), evaluated in log-space ($a \log w + b \log(1-w) - \log\gamma - \texttt{betaln}(a,b)$, exponentiated at the end) to avoid under/overflow when w is close to 0 or 1. Verified against a central-difference derivative of `GB2LogLink.cdf` itself (independent of this derivation) at an arbitrary parameter point: relative error 8.6×10^{-11} – floating-point precision, not an approximation.